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Assessment of termite infestation and its impact on structures and agriculture in Madhepura, Bihar

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Abstract- Termites are among the most destructive pests worldwide, causing significant damage to wooden structures and agricultural crops. This study assesses termite infestation patterns and their impact on structures and agriculture in Madhepura, Bihar. A systematic survey was conducted over six months (January, 2023 to June, 2023) across urban and rural areas, covering residential buildings, agricultural fields, and uncultivated lands. Data collection involved visual inspections, baiting methods, soil sampling, and laboratory identification of termite species. Results revealed that 76% of the surveyed sites showed termite activity, with the highest infestation recorded in residential buildings (85%), followed by agricultural fields (70%) and forested areas (60%). Three major termite species were identified: *Odontotermes obesus* (subterranean termites), *Cryptotermes brevis* (drywood termites), and *Neotermes bosei* (dampwood termites). Subterranean termites were the most prevalent, particularly in moist soil conditions, contributing to severe structural damage and crop losses. Sugarcane fields suffered a 25-30% loss, while wheat and maize recorded a 20-25% yield reduction due to termite feeding on roots and stems. A correlation analysis demonstrated that termite activity was highest in areas with high moisture content, organic-rich soils, and inadequate drainage, particularly during the monsoon season. The study highlights the urgent need for integrated pest management (IPM) strategies to mitigate termite damage. Recommendations include pre-construction soil treatments with termiticides, borate-based wood treatments, improved moisture control, crop rotation with termite-resistant plants, and the use of biological control agents such as entomopathogenic fungi and nematodes. Additionally, community awareness programs and periodic monitoring are essential for early detection and sustainable management. This research provides critical insights into the distribution and impact of termites in Madhepura, contributing valuable data for developing region-specific termite management policies. Future research should focus on long-term infestation monitoring, eco-friendly control alternatives, and enhancing resistance in crops and building materials against termite attacks.

Key words: Termite infestation, structural damage, agricultural loss, subterranean termites, pest management,

INTRODUCTION

Termites are social insects belonging to the order Blattodea and the infraorder Isoptera. They are often called "silent destroyers" because of their ability to chew through wood, flooring, and even wallpaper undetected. These small but highly organized insects play a crucial role in ecosystems by decomposing plant material and recycling nutrients. However, some species are notorious for causing extensive damage to human structures and agricultural crops. There are over 2,700 known termite species, categorized into three main types: subterranean, drywood, and dampwood termites. Subterranean termites live in underground colonies and build mud tubes to access food sources, making them the most destructive type. Drywood termites, as their name suggests, infest dry wood without needing contact with soil, while dampwood termites prefer

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moist and decaying wood. Termites have a complex social structure consisting of three primary castes: workers, soldiers, and reproductives. Workers are responsible for foraging, feeding, and maintaining the nest, while soldiers defend the colony against predators, primarily ants. The reproductives include the king and queen, who ensure the colony's growth, and alates (winged termites), which leave the nest to establish new colonies.[1,2] While termites are often associated with destruction, they also provide ecological benefits. They enhance soil fertility by breaking down organic matter, promote plant growth, and serve as a food source for various animals. However, their destructive impact on human structures results in billions of dollars in damages worldwide annually. Effective termite management includes chemical treatments, baiting systems, and preventive construction practices. Understanding termites' behavior, ecology, and impact helps in developing sustainable control strategies, balancing their ecological role with the need to protect human properties.[3,4]

A termite survey is crucial for understanding the extent of termite infestation, assessing potential damage, and implementing effective control measures. Termites are destructive pests that can silently weaken wooden structures, agricultural crops, and forests, leading to significant economic losses.

Conducting regular surveys helps in early detection and strategic management.

1. Early Detection and Prevention

Termites often remain hidden within structures, making it difficult to notice infestations until severe damage has occurred. Surveys help in identifying termite presence at an early stage, allowing preventive measures to be taken before the damage becomes extensive.

2. Protecting Buildings and Infrastructure

Buildings made of wood or containing wooden components are highly vulnerable to termite attacks. A survey helps in mapping infestation-prone areas and taking corrective actions such as soil treatment, chemical barriers, or using termite-resistant materials in construction.

3. Agricultural and Forestry Protection

Termites pose a major threat to crops like sugarcane, wheat, and maize, as well as fruit trees and forest plantations. Surveys help in assessing their impact on agriculture and developing sustainable control strategies to protect food production and biodiversity.

4. Economic Savings

Termite damage leads to billions of dollars in repair costs worldwide. By identifying and addressing infestations early through surveys, property owners and farmers can save money on repairs and pest control treatments.

5. Ecological Balance and Sustainable Control

While termites are known for their destructive nature, they also play a key role in breaking down organic matter and enriching soil. A termite survey helps in distinguishing between harmful infestations and beneficial termite populations, enabling eco-friendly pest management practices.

6. Scientific Research and Policy Formulation

Surveys provide data for entomologists, ecologists, and government agencies to study termite behavior, spread, and control methods. This research supports the development of effective policies for termite management in urban and rural areas.

MATERIALS & METHODS

1. Study Area

The survey was conducted in Madhepura district, Bihar, covering both urban and rural regions. The selected sites included: Residential buildings (houses, schools, government offices) Agricultural fields (sugarcane, wheat, maize, and other termite-prone crops) Forested and uncultivated lands

2. Survey Duration

The study was carried out over a period of six months (from January, 2023 to June, 2023), covering different seasons to understand termite activity variations.

3. Materials Used Field equipment

Hand augers, soil samplers, and collection vials. Baiting traps, Cardboard and wooden stakes placed in different locations to attract termites. Preservation materials, Ethanol (70%) for specimen preservation Microscopes and identification keys for species-level identification GPS device and maps for location marking. Digital camera for photographic documentation

4. Sampling Methodology

Visual Inspection Structural buildings and wooden components were inspected for termite damage (mud tubes, wood hollowness, and fecal pellets). Soil samples were collected near foundations and agricultural fields. Baiting Method Wooden stakes and cardboard traps were buried

in the soil at a depth of 10-15 cm and checked at weekly intervals for termite activity. Hand-Collection & Soil Sampling Live termites were collected using hand tools from infested areas, stored in ethanol, and transported to the laboratory for identification.

5. Identification of Termites

Collected samples were examined under a stereomicroscope and identified based on morphological features such as head shape, mandibles, and wing venation using standard taxonomic keys. The species were categorized as subterranean, drywood, or dampwood termites based on habitat and feeding habits.

6. Data Analysis

Infestation rates were calculated as the percentage of structures or fields affected. Species distribution was mapped using GIS tools. Correlation analysis was performed to assess relationships between termite infestation and environmental factors (humidity, temperature, soil type).

7. Control Measures and Recommendations

Based on survey findings, possible termite control strategies, including biological, chemical, and cultural control methods, were evaluated for sustainable management in the region. This methodology ensured a comprehensive assessment of termite infestation and its impact on structures and agriculture in Madhepura, Bihar.

RESULTS & DISCUSSION

Termite Infestation Patterns in Madhepura, Bihar. The survey revealed widespread termite infestation across different locations in Madhepura. Out of the 50 surveyed sites, 38 (76%) showed signs of termite presence. The highest infestation was recorded in residential buildings (85%), followed by agricultural fields (70%) and forest/ uncultivated land (60%). The intensity of infestation varied across locations, with higher occurrences in areas with moist soil, poor drainage, and wooden structures in contact with the ground. Three major termite species were identified: *Odontotermes obesus* (Subterranean termites) most destructive, found in 65% of surveyed locations, particularly in moist soils near building foundations and agricultural fields. *Cryptotermes brevis* (Drywood termites) found in 25% of the sampled buildings, especially in wooden furniture and structural components. *Neotermes bosei* (Dampwood termites) recorded in 10% of cases, mainly in decaying wood in forested areas. Microscopic examination confirmed the distinct morphological characteristics of each species, such as head size, mandibles, and wing venation patterns. Subterranean termites were the most dominant due to favorable soil conditions and the availability of cellulose-based food sources.[5,6]

Significant termite activity was observed in wooden beams, door frames, and furniture in buildings. Mud tubes were detected in 60% of structures, indicating active subterranean termite presence. Some houses showed severe weakening of structural wood, raising safety concerns. Sugarcane fields experienced 25-30% crop loss due to subterranean termite feeding on roots and stems. Wheat and maize crops showed 20-25% yield reduction in affected areas. Infestation was more prevalent in fields with high organic matter and poor soil drainage.[7,8] A correlation analysis between temperature, humidity, and soil type revealed that: Higher termite activity was recorded in moist soil with organic matter content above 3%.

Termite infestation was more prominent in humid conditions (above 60% relative humidity), explaining increased activity during the monsoon season. In dry conditions, drywood termites showed resilience, but overall termite movement decreased.

Based on the findings, several termite management strategies are recommended:

1. For Buildings-

Use of pre-treatment soil barriers with termiticides during construction. Application of borate-based wood treatments for increased resistance. Improved drainage and moisture control around buildings.

2. For Agriculture-

Crop rotation with termite-resistant plants to reduce infestation. Use of biological control agents like nematodes and fungal pathogens. Soil amendment with neem-based biopesticides for eco-friendly termite management.

3. Community Awareness and Regular Inspections-

Training local farmers and homeowners on early detection techniques. Periodic termite surveys to monitor infestation trends and evaluate control effectiveness.

CONCLUSION

This study underscores the widespread impact of termite infestation on both structures and agriculture in Madhepura, Bihar. The high infestation rates in residential buildings and agricultural fields highlight the urgent need

for proactive management strategies. Among the identified species, subterranean termites were the most destructive, particularly in moisture-rich environments, leading to significant structural degradation and crop losses. Given the severity of the problem, an integrated pest management (IPM) approach is essential. Preventive measures such as soil treatment with termiticides, moisture control, and borate-based wood treatments can enhance building resilience. In agriculture, sustainable strategies like crop rotation with termite-resistant plants, biological control agents, and organic soil amendments should be prioritized to mitigate termite-related losses. Community awareness, periodic monitoring, and research-driven innovations in termite-resistant construction materials and eco-friendly pest control methods will play a crucial role in long-term management. Future research should focus on advanced monitoring techniques, the development of bio-rational pesticides, and enhancing genetic resistance in crops to termite attacks. A coordinated effort between researchers, policymakers, and local communities is imperative to mitigate termite damage effectively and sustainably in the region.

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